

# Deep Reinforcement Learning for irrigation optimization using a digital twin of crop-soil dynamics

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**Abstract**—Irrigation management is a key challenge in modern agriculture, where water scarcity and climate variability demand smarter, data-driven decision systems. We propose a deep reinforcement learning (DRL) framework for irrigation scheduling based on a digital twin of crop-soil dynamics. Synthetic multi-year weather scenarios are generated from historical climate data via a KNN bootstrap resampling procedure, ensuring broad climatic coverage across training conditions. Daily soil water tension is estimated from meteorological inputs and field measurements using a linear autoregressive model with exogenous variables (ARX), while AquaCrop handles crop biomass simulation under varying irrigation events. A DRL agent is then trained within this simulated environment using the Proximal Policy Optimization (PPO) algorithm to learn an irrigation policy balancing crop productivity against water consumption. The framework is assessed on field data collected from a vineyard in the Val d’Adige region of Trentino, Italy, over the 2023 and 2024 irrigation seasons. Preliminary results show that the synthetic weather data reproduces the statistical properties of the historical climate record, the ARX soil tension model achieves an  $R^2$  of 0.956 on held-out test data, and the DRL component provides initial evidence that irrigation strategies broadly consistent with validated fuzzy decision support systems can be learned without requiring expert-defined rules.

**Index Terms**—Reinforcement learning, AquaCrop, IoT-based agriculture, irrigation management, precision agriculture

## I. INTRODUCTION

The growing pressure on global freshwater resources, together with the increasing unpredictability of climate change, has made irrigation management one of the most urgent challenges in modern agriculture. Precision agriculture and the Internet of Things (IoT) have opened the door to data-driven Decision Support Systems (DSS) that aim to improve water use efficiency without sacrificing crop productivity. For irrigation scheduling, these systems have long relied on heuristic practices or fuzzy logic controllers [1], [2]. These methods work well when conditions are known in advance, but they carry an intrinsic limitation: the irrigation strategy must be defined a priori, encoded as explicit rules or fixed thresholds. When conditions deviate from what was anticipated at design time, the system has no mechanism to adapt.

Digital Twins have recently gained traction as a way to address this gap [3]. A Digital Twin maintains a virtual replica of the physical field, providing a simulation environment where strategies can be tested and refined without putting real crops at risk. This is where Reinforcement Learning (RL)

becomes particularly appealing [4], [5]. Rather than following a predefined strategy, an RL agent learns by interacting with the environment, gradually improving its decisions as it is exposed to varying agronomic and climatic conditions. No rules need to be written in advance. The application of Deep Reinforcement Learning (DRL) to irrigation scheduling has grown considerably since Yang et al. [6] first demonstrated its feasibility, with subsequent studies addressing a variety of crops and field conditions [7], [8]. Kelly et al. [9], for instance, trained a DRL agent within the AquaCrop crop simulator [10], showing that it outperforms conventional heuristics under low rainfall conditions. Saikai et al. [11] reached similar conclusions in a different setting, training an agent on high-dimensional sensor data within the APSIM simulator and consistently outperforming rule-based approaches.

Despite these significant advancements, a major challenge remains. Training a robust RL agent requires exposure to a wide and diverse range of conditions, but historical weather records are limited and short-term field observations rarely capture the variability needed to generalize beyond the training period. Data scarcity, in other words, is not a minor inconvenience, but is one of the core obstacles to deploying DRL-based irrigation systems in practice.

This paper presents a DRL framework for vineyard irrigation scheduling, developed and validated within an agricultural consortium located in the Roverè della Luna municipality (hereafter RDL), Val d’Adige region of Trentino, Italy. The framework is built on a digital twin of the local crop-soil dynamics, designed to address data scarcity and to bridge the gap between physics-based crop models and real-time field sensor measurements. To address this limitation, we introduce a K-Nearest Neighbors (KNN) bootstrap resampling procedure [12] that generates synthetic multi-year weather sequences from historical climate records, substantially extending the range of scenarios the agent encounters during training. Soil water tension is estimated daily by a linear autoregressive model with exogenous variables (ARX), selected for its reliability in low-data regimes over more complex alternatives. Crop growth under varying irrigation strategies is simulated using the FAO AquaCrop model [10]. A DRL agent is then trained within this environment using Proximal Policy Optimization (PPO) [13], learning a policy that balances crop productivity with water consumption. These four components

together constitute a digital twin of the crop-soil system.

The remainder of this paper is organized as follows. Sec. II describes the proposed methodology, detailing the four components of the digital twin architecture. Sec. III presents the simulation pipeline and the agent training procedure. Finally, Sec. IV reports preliminary validation results and discusses ongoing and future work.

## II. PROPOSED METHODOLOGY

The proposed methodology consists of four interconnected components, as illustrated in Figure 1. First, a KNN bootstrap resampling procedure generates synthetic long-term weather scenarios from historical climate data. Second, an ARX model estimates the daily soil water tension from meteorological variables and irrigation records. Next, the AquaCrop model simulates the daily accumulation of biomass given a sequence of irrigation events and meteorological inputs. Finally, a DRL agent is trained in this simulated environment using the PPO algorithm to learn an irrigation policy that maximizes productivity while minimizing water consumption. Together, these components constitute a digital twin of crop-soil dynamics, enabling the identification of an optimal irrigation strategy across a wide range of climatic scenarios. In the remainder of this section, we describe each component in detail.

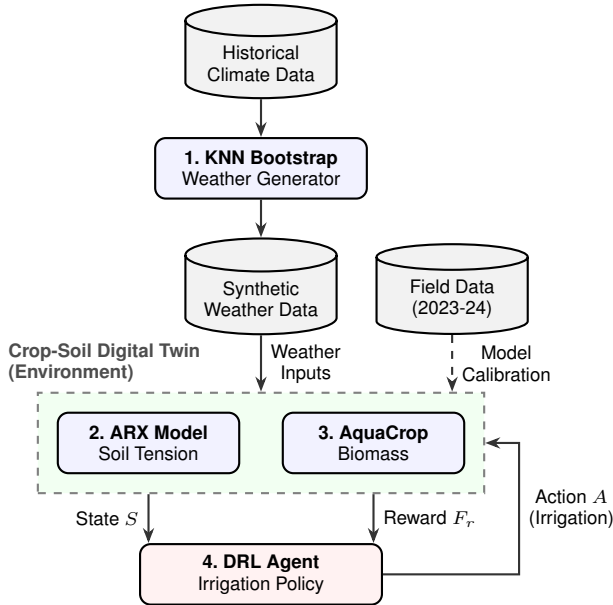


Fig. 1. Overall architecture of the proposed framework. Rectangles represent processing components; cylinders represent data sources.

### A. Climate Data Acquisition and Scenario Generation

Let  $\mathcal{D} = \{(\mathbf{x}_i, d_i)\}_{i=1}^N$  denote the historical dataset derived from OpenMeteo, where  $N$  is the total number of observations,  $\mathbf{x}_i \in \mathbb{R}^p$  is the feature vector at time step  $i$  including the current meteorological observations and their lagged values over the preceding 3 days, and  $d_i \in \{1, \dots, 366\}$  is the corresponding day of the year.

Historical data were retrieved for the coordinates of RDL over the 30-year period spanning from January 1, 1990 to December 31, 2019. This temporal window was chosen for two reasons: it provides a sufficiently large number of observations to ensure robustness in the weather simulation, and it corresponds to a defined climate reference period associated with the conditions observed in recent years [14]. All variables are aggregated on a daily level and include FAO reference evapotranspiration ( $ET_0$ ), cumulative precipitation ( $P^{\text{tot}}$ ), mean relative humidity at 2 meters (m) above ground (RH), minimum and maximum temperature at 2 m ( $T^{\text{min}}$ ,  $T^{\text{max}}$ ), accumulated shortwave radiation (SR), and mean wind speed at 10 m (W). These variables were selected because they represent the primary meteorological drivers of soil water tension and, by extension, of soil moisture dynamics.

The procedure is formalized in Algorithm 1, which describes how, given a window size  $w \in \mathbb{N}$  and a number of neighbors  $k \in \mathbb{N}$ , a synthetic weather dataset of length  $T$  is generated by iteratively sampling successor states.

The seasonal window  $\mathcal{W}(d_t, w)$  restricts the neighbor search to historically comparable days, so that the simulated sequence respects the climatological structure of the observed record. The use of successor records rather than the sampled ones themselves ensures temporal continuity and realistic day-to-day transitions in the generated weather data. At each iteration, the lag values of all variables are updated to maintain internal consistency throughout the generated trajectory. Through repeated application of the KNN bootstrap procedure, the historical record was expanded into a synthetic dataset of 1000 simulated years.

### B. Soil Tension Estimation via an Autoregressive Model

Soil water tension at 30 cm depth was estimated using an ARX model trained on data collected during the irrigation seasons of 2023 and 2024 at RDL. The feature set includes the meteorological variables retrieved from OpenMeteo described in the previous section, along with their lagged values over the preceding 3 days, daily irrigation records in mm, and soil tension measurements in mbar collected from 9 tensiometers and aggregated to a daily mean. The inclusion of lagged soil tension values as predictors gives the model its autoregressive structure, allowing it to capture the temporal dynamics of soil moisture depletion and recharge.

The ARX model was selected for its simplicity and interpretability, and because linear models tend to generalize well in low-data regimes, a relevant consideration given the limited number of irrigation seasons available. To verify that this choice was not detrimental to predictive accuracy, its performance was compared against more sophisticated models. The comparison was carried out via expanding window cross-validation: starting from an initial training window of 400 days, the model is retrained at each step on all data available up to the current date and evaluated on the following day, progressively expanding the training set as new observations become available. This procedure provides a realistic estimate of out-of-sample performance under the temporal structure

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**Algorithm 1** KNN Bootstrap Resampling for Weather Data

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- 1: **Input:** Historical dataset  $\mathcal{D} = \{(\mathbf{x}_i, d_i)\}_{i=1}^N$ ; window size  $w$ ; number of neighbors  $k$ ; simulation horizon  $T$
  - 2: **Output:** Simulated weather dataset  $\{\mathbf{x}_t\}_{t=1}^T$
  - 3: **for** each time step  $t = 1$  to  $T - 1$  **do**
  - 4:   **Step 1: Seasonal window selection**
  - 5:   Define the set of historically admissible candidates:  
     $\mathcal{W}(d_t, w) = \{i \in \{1, \dots, N\} \mid |d_i - d_t| \leq w\}$
  - 6:   **Step 2: Nearest neighbor search**
  - 7:   For each candidate  $i \in \mathcal{W}(d_t, w)$ , compute the Euclidean distance:  
     $\delta_i = \|\mathbf{x}_t - \mathbf{x}_i\|_2$
  - 8:   Retain the  $k$  candidates with the smallest distance:  
     $\mathcal{N}_k(\mathbf{x}_t) = \arg \min_{i \in \mathcal{W}(d_t, w)} \delta_i$
  - 9:   **Step 3: Weighted random sampling**
  - 10:   Assign to each neighbor  $i \in \mathcal{N}_k(\mathbf{x}_t)$  an inverse distance weight:  
     $\tilde{p}_i = \frac{1/\delta_i}{\sum_{j \in \mathcal{N}_k(\mathbf{x}_t)} 1/\delta_j}$
  - 11:   Sample one neighbor  $i^*$  at random according to weights  $\{\tilde{p}_i\}$ :  
     $P(i^* = i) = \tilde{p}_i, \quad \forall i \in \mathcal{N}_k(\mathbf{x}_t)$
  - 12:   **Step 4: Successor extraction**
  - 13:   Set the next state and update the day of the year using the immediate historical successor:  
     $\mathbf{x}_{t+1} = \mathbf{x}_{i^*+1}$   
     $d_{t+1} = d_{i^*+1}$
  - 14:   **Step 5: Lag update**
  - 15:   Update the lagged values in  $\mathbf{x}_{t+1}$  by replacing the historical lags with the previously simulated states  
     $\{\mathbf{x}_t, \mathbf{x}_{t-1}, \mathbf{x}_{t-2}\}$
  - 16: **end for**
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of the data. Final performance of the selected model is then reported on the test set.

The dataset spans from April 12, 2023 to October 24, 2024, for a total of 562 daily observations. The training and validation set covers the period from April 12, 2023 to July 31, 2024 (477 samples), while the test set covers the period from August 1 to October 24, 2024 (85 samples). This split was designed to ensure that both partitions contain periods with and without irrigation events, so as to cover the full range of agronomic conditions encountered during the growing season. Since the ARX model has no hyperparameters to tune, it was trained directly on the full training and validation period to obtain the final parameters, without requiring any model selection procedure.

### C. The AquaCrop Simulation Environment

AquaCrop is a crop water productivity model developed by the Food and Agriculture Organization (FAO) of the United Nations, designed to simulate crop yield and water use under varying management and environmental conditions [10]. In this work, AquaCrop was configured to represent the specific agronomic and pedological conditions of RDL. Soil parameters were derived from local field measurements conducted in the study area, while crop parameters were calibrated for the Lagrein grapevine variety by combining domain knowledge provided by local agronomists with values reported in the literature for comparable cultivars [15].

The model operates at a daily time step and receives as input  $ET_0$ ,  $T^{\min}$ ,  $T^{\max}$ , and  $P^{\text{tot}}$ , consistent with the variables retrieved from OpenMeteo. Given a sequence of daily irrigation decisions  $\{I_t\}_{t=1}^N$ , AquaCrop yields daily estimates of aboveground biomass  $B_t$  and potential biomass  $B_t^{\max}$ , where the latter represents the theoretical maximum accumulation under non-limiting water conditions and serves as the normalization reference in the reward function (2).

### D. Reinforcement Learning Formulation

The irrigation scheduling problem is formulated as a Markov Decision Process (MDP)  $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R} \rangle$ , where  $\mathcal{S}$  is the state space,  $\mathcal{A}$  is the action space,  $\mathcal{T}$  is the transition dynamics, and  $\mathcal{R}$  is the reward function. The goal of the agent is to learn a policy  $\pi_\theta : \mathcal{S} \rightarrow \mathcal{A}$  that maximizes the expected cumulative reward over a growing season.

**State space.** At each day  $t$ , the state vector is defined as:

$$\mathbf{S}_t = \left( \bar{\phi}_t, \hat{\phi}_{t+1}, \hat{P}_{t,3}^{\text{tot}}, \hat{T}_{t,3}^{\max} \right) \quad (1)$$

where  $\bar{\phi}_t$  is the average soil water tension estimated by the ARX model at day  $t$ ;  $\hat{\phi}_{t+1}$  is the predicted average soil tension for the following day under the assumption of no irrigation;  $\hat{P}_{t,3}^{\text{tot}}$  and  $\hat{T}_{t,3}^{\max}$  are the forecasted cumulative precipitation and maximum temperature over the next 3 days, respectively. This state representation provides the agent with both the current soil water status and short-term meteorological information, enabling anticipatory irrigation decisions.

**Action space.** The action  $a_t \in [0, I_{\max}]$  represents the amount of water (in mm) to be delivered on day  $t$ , where  $I_{\max}$  is the maximum irrigation volume deliverable in a single day. The action space is continuous, and the policy  $\pi_\theta$  outputs a distribution over this interval from which the irrigation quantity is sampled during training and taken deterministically at inference time.

**Reward function.** The reward  $F_r$  is designed to jointly incentivize crop productivity, water use efficiency, and maintenance of optimal soil water conditions:

$$F_r = w_1 \cdot \frac{B}{B_{\max}} - w_2 \cdot \frac{I}{I_{\max}} - w_3 \cdot f(\phi) \quad (2)$$

where  $B$  and  $B_{\max}$  are the daily biomass and potential biomass estimated by AquaCrop,  $I$  is the irrigation action suggested by the agent,  $I_{\max}$  is the maximum irrigation

volume deliverable in a single day, and  $f(\phi)$  is a penalty term penalizing deviations of the soil tension from the agronomically optimal range. The penalty function is defined as:

$$f(\phi) = \begin{cases} 0 & \text{if } \phi_{\min} \leq \phi \leq \phi_{\max} \\ \frac{\phi_{\min} - \phi}{\phi_{\text{scale}}} & \text{if } \phi < \phi_{\min} \\ \frac{\phi - \phi_{\max}}{\phi_{\text{scale}}} & \text{if } \phi > \phi_{\max} \end{cases} \quad (3)$$

where  $\phi_{\min}$  and  $\phi_{\max}$  define the optimal tensiometric range, and  $\phi_{\text{scale}}$  is a normalization factor. For the grapevine crop considered in this study, these are set to  $\phi_{\min} = 200$  mbar,  $\phi_{\max} = 600$  mbar, and  $\phi_{\text{scale}} = 200$  mbar. The weights  $w_1, w_2, w_3$  allow tuning the trade-off between biomass maximization, water saving, and soil tension regulation; in this work, all weights are set to  $w_1 = w_2 = w_3 = 1$ . A systematic tuning of these weights is beyond the scope of this paper and is left as future work.

**Optimization algorithm.** The policy  $\pi_\theta$  is optimized using Proximal Policy Optimization (PPO) [13], a state-of-the-art actor-critic algorithm that constrains policy updates via a clipped surrogate objective, ensuring stable and sample-efficient learning. The full training procedure is detailed in Algorithm 2.

### III. SIMULATION PIPELINE

The training loop, detailed in Algorithm 2, coordinates the interaction between the components described in Section II: the ARX soil tension model, the AquaCrop crop simulator, and the DRL agent. At each time step  $t$  within a simulated season, the ARX model first predicts the next-day soil tension  $\hat{\phi}_{t+1}$  under the assumption of no irrigation, providing the agent with a counterfactual baseline. The agent then constructs the state vector  $S_t$  and outputs an irrigation recommendation  $\hat{I}_{t+1}$ . If irrigation is prescribed, the soil tension estimate is updated accordingly; otherwise, the baseline prediction is retained. AquaCrop is subsequently queried to estimate the resulting biomass  $\hat{B}_{t+1}$ , and the reward  $F_r$  is computed as defined in Equation (2). The transition  $(S_t, \hat{I}_{t+1}, F_r, S_{t+1})$  is stored in a rollout buffer, which is used at the end of each season to update the policy via Generalized Advantage Estimation (GAE) and the PPO algorithm.

### IV. PRELIMINARY RESULTS AND CONCLUSION

Preliminary validation of the individual pipeline components yields encouraging results. The KNN bootstrap resampling procedure successfully reproduces the statistical properties of the historical climate record. Validation was carried out by comparing the synthetic and historical data across three dimensions: marginal distributions of each meteorological variable, pairwise correlations among variables, and the annual cycle of mean values. In all three cases, the synthetic sequences closely match the historical record, suggesting that the resampling procedure preserves both the univariate behavior and the multivariate dependence structure of the original data,

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### Algorithm 2 RL loop for Irrigation Scheduling

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1: Input: Simulated weather dataset  $\{\mathbf{x}_t\}_{t=1}^T$ ; ARX model
    $f(\cdot)$ ; AquaCrop simulator; reward function
   weights  $(w_1, w_2, w_3)$ 
2: Output: Trained irrigation policy  $\pi_{\theta^*}$ 

3: for each simulated weather season  $y = 1$  to  $Y$  do
4:   for each day  $t = 1$  to  $N$  do
5:     Step 1: Simulate soil tension with ARX model
6:     Predict  $\hat{\phi}_{t+1}$  assuming no irrigation ( $I_{t+1} = 0$ ):
        $\hat{\phi}_{t+1} = f(\mathbf{x}_t, I_{t+1} = 0)$ 
7:     Step 2: Suggest irrigation with the DRL agent
8:     Construct the state vector and compute the
       recommendation:
        $S_t = (\bar{\phi}_t, \hat{\phi}_{t+1}, \hat{P}_{t,3}^{\text{tot}}, \hat{T}_{t,3}^{\text{max}})$ 
        $\hat{I}_{t+1} = \pi_\theta(S_t)$ 
9:     Step 3: Update soil tension and biomass
10:    if  $\hat{I}_{t+1} \neq 0$  then
11:      Update  $\hat{\phi}_{t+1}$  to account for the irrigation  $\hat{I}_{t+1}$ 
12:    else
13:      Proceed without irrigation; keep  $\hat{\phi}_{t+1}$ 
14:    end if
15:    Predict the biomass  $\hat{B}_{t+1}$  with AquaCrop
16:    Step 4: Compute the reward
        $F_r = w_1 \cdot \frac{\hat{B}_{t+1}}{\hat{B}_{\text{max}}} - w_2 \cdot \frac{\hat{I}_{t+1}}{\hat{I}_{\text{max}}} - w_3 \cdot f(\phi)$ 
17:    Store transition  $(S_t, \hat{I}_{t+1}, F_r, S_{t+1})$  in the buffer
18:  end for
19:  Step 5: Update policy
20:  Compute advantage estimates via Generalized Ad-
    vantage Estimation (GAE)
21:  Optimize  $\pi_\theta$  via the PPO algorithm
22:  Clear the buffer
23: end for

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as well as its seasonal patterns. The ARX soil tension model attains an RMSE of 60.6 mbar, a MAE of 31.5 mbar, and an  $R^2$  of 0.956 on the test set, confirming its ability to generalize across the full range of agronomic conditions encountered during the growing season. Notably, these results compare favorably with more expressive tree-based models such as XGBoost, as well as with a persistence baseline that predicts tomorrow's tension as equal to today's observed value. This suggests that the linear autoregressive structure is sufficient to capture the dominant dynamics of soil moisture and that added model complexity does not translate into improved predictive accuracy given the limited data available. While these components are not the primary contribution of this work, their accuracy is a prerequisite for the reliability of the simulation environment in which the DRL agent is trained.

A full quantitative comparison across models will be reported in the final paper.

Preliminary results on the DRL component support the feasibility of the proposed framework. When evaluated in simulation over the 1000-year synthetic weather dataset, the learned policy produces irrigation strategies that maintain a reasonable trade-off between biomass preservation and water use across heterogeneous climatic scenarios. As a reference, its behavior is compared against the fuzzy decision support systems developed in our previous work [1], [2], which were validated on field data collected at the same site. At the current stage, the evidence indicates that the DRL agent can learn irrigation policies that exhibit behavior consistent with the established agronomic rationale encoded in those rule-based systems, while requiring no expert-defined control rules. A full quantitative comparison in terms of seasonal water use and crop productivity is part of the ongoing validation.

In conclusion, this paper has presented a DRL framework for irrigation scheduling built upon a digital twin of the crop-soil system, integrating synthetic weather generation, autoregressive soil tension estimation, and crop simulation. The proposed approach is designed to operate without requiring real-time sensor data during training, relying instead on historically grounded simulations to expose the agent to a broad spectrum of conditions.

This extended abstract presents the proposed framework, the validation of its main simulation components, and initial evidence on the learned irrigation policy. A complete quantitative assessment on the 2023-2024 RDL field data, including seasonal water savings and cumulative reward comparisons against actual irrigation records, is left to the post-acceptance extended version.

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